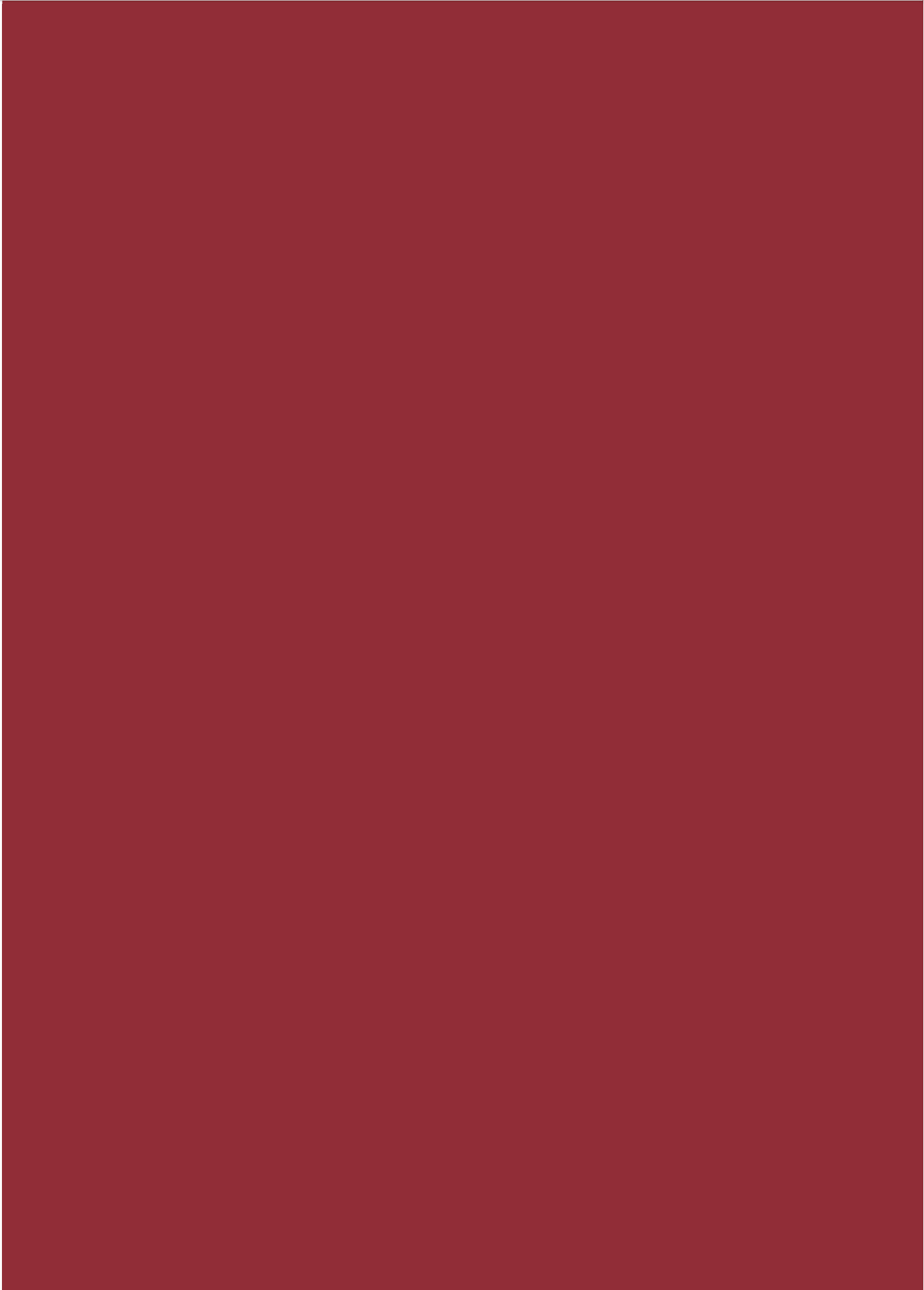


← OB - Account Inquiry APIs for client

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Implementation Guidelines for Client

Open Banking - Account Inquiry

Application Programming Interface Specification

Version 1.0.4

← OB - Account Inquiry APIs for client

Version	Date	Prepared By	Description
1.0.0	08/04/2024	Vortey VANN	Initial version
1.0.1	05/06/2024	Vortey VANN	-In the API response body, under the 'account type' field, list 'LOAN' as an allowed value. - Modify the API QR Inquiry to handle remittance and merchant QR code formats, returning relevant account data in the response.
1.0.2	28/01/2025	Phoeun Sorya Phirak Tepisandap	- Modify the response data for API QR Inquiry - Modify the unexpected error code - Updated keyname for Merchant Category Code in APIs Merchant Inquiry and QR Inquiry - Add length constraint for paymentRef
1.0.3	20/06/2025	Phoeun Sorya	Modify the response data for API Account Inquiry
1.0.4	05/08/2025	Phoeun Sorya	Updated on errorCode data type

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← OB - Account Inquiry APIs for client

Introduction

Overview

Innovating the Open Banking landscape, the Account Inquiry Module for Open Banking addresses the current limitation of real-time account information retrieval especially for KHQR payment. Clients/Providers will be empowered to access real-time account data for their users, reducing the risk of fraud/scams. This will enhance user experience, drive innovation, and strengthen financial literacy within the Open Banking ecosystem.

Purpose

This document describes the detailed specification API of the Account Inquiry Module as a guideline for all Account Inquiry Module members to access Open Banking and account inquiry. The expected readers are the NBC technical team and the third-party technical team. This can be used as a reference for any interest related to the API mentioned in this document.

Scope

This document contains the complete description of the API specification including the endpoint, status code, request, and response sample.

Overall API

Name	Method	URL
Authentication	POST	{{protocol}}://{{port}}/v1/authenticate
Account Inquiry	POST	{{protocol}}://{{host}}/api/v1/account_inquiry/account
Account Inquiry Batch	POST	{{protocol}}://{{host}}/api/v1/account_inquiry/account_inquiry_batch
Merchant Inquiry	POST	{{protocol}}://{{host}}/api/v1/account_inquiry/merchant_inquiry
QR Inquiry	POST	{{protocol}}://{{host}}/api/v1/account_inquiry/qr_inquiry

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This sequence diagram has 4 Participants:

- **Client (C)**: The entity initiating the Account Inquiry Module request.
- **Open Banking (OB)**: The platform facilitating access to financial data through APIs.
- **Account Inquiry Module (AI)**: The service processing account information requests.
- **Provider's Account Inquiry Module (P)**: The specific service within the Provider's system that handles account inquiries.

Pre- condition

1. Client Authentication:

- The Client sends an authentication request (**API/authentication**) to the Open Banking provider.
- The Open Banking provider receives and validates the request.
 - If valid credentials: The provider issues an access token (JWT) to the client.
 - If invalid credentials: An error message is returned.

2. Provider Authentication:

- The Open Banking provider sends an internal authentication request (**API/authentication**) to its own account inquiry service (Provider's Account Inquiry).
- The Provider's Account Inquiry receives and validates the request.
 - If valid credentials: An access token (JWT) is issued internally.
 - If invalid credentials: An error message is returned.

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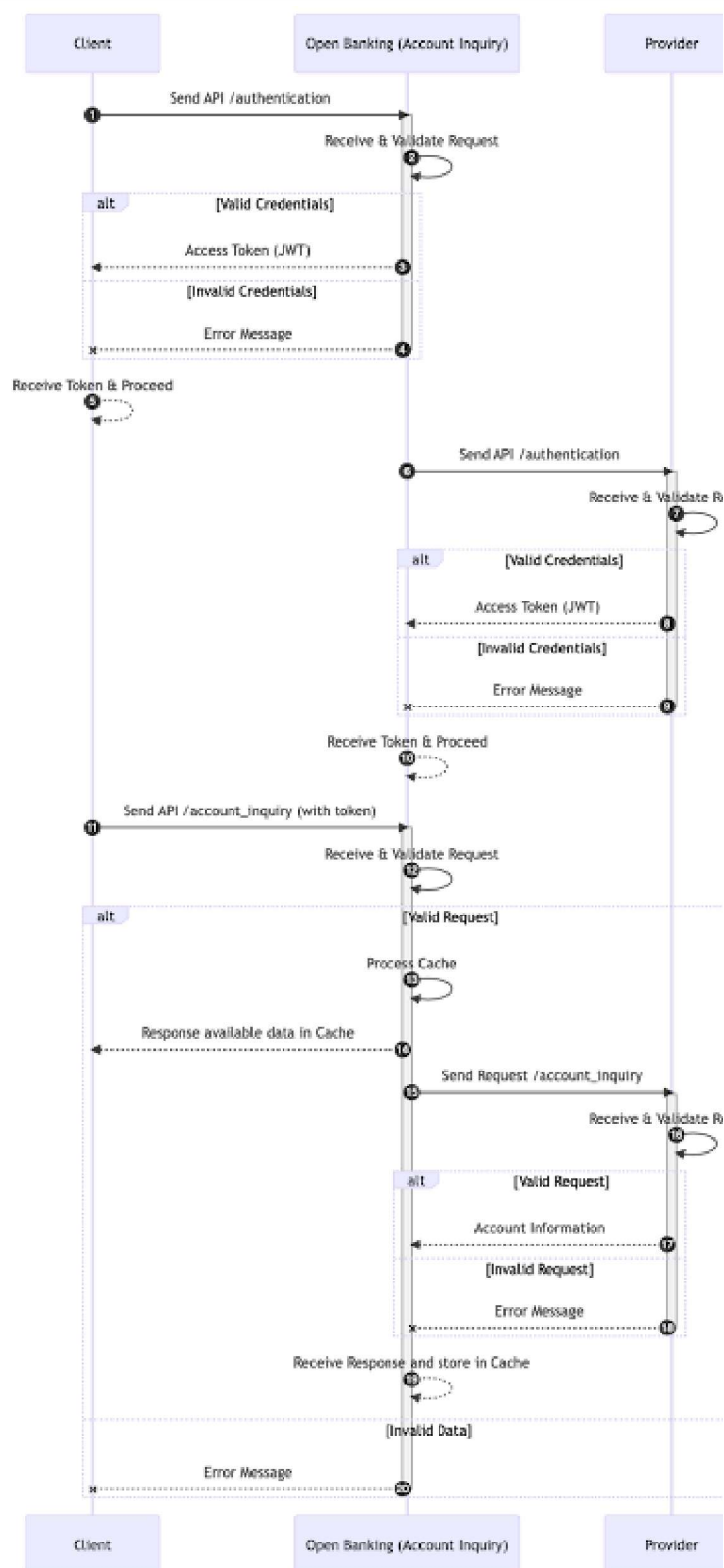


figure 1: Sequence diagram - account inquiry

1. [Client Authentication](#)
2. [Provider Authentication](#)
3. Account Inquiry with Token:

← OB - Account Inquiry APIs for client

- The provider validates the request.
 - If valid request: The process proceeds to check the cache.
 - If invalid request: An error message is returned.

4. Cache Check:

- The provider checks its cache to see if the requested account information is already available.
- If the data is in the cache, the provider retrieves the information and sends it back to the client.

5. Account Information Retrieval (If not in cache):

- If the data is not in the cache, the provider sends a request to the account inquiry service to fetch the information.
 - The account inquiry service receives and validates the request.
 - If the request is valid, the account information is returned.
 - If the request is invalid, an error message is returned.

6. Cache Storage and Response:

- The provider receives the account information (either from the cache or the account inquiry service), stores it in the cache, and sends the response back to the client.

Main Flow - Account Inquiry Batch

← OB - Account Inquiry APIs for client

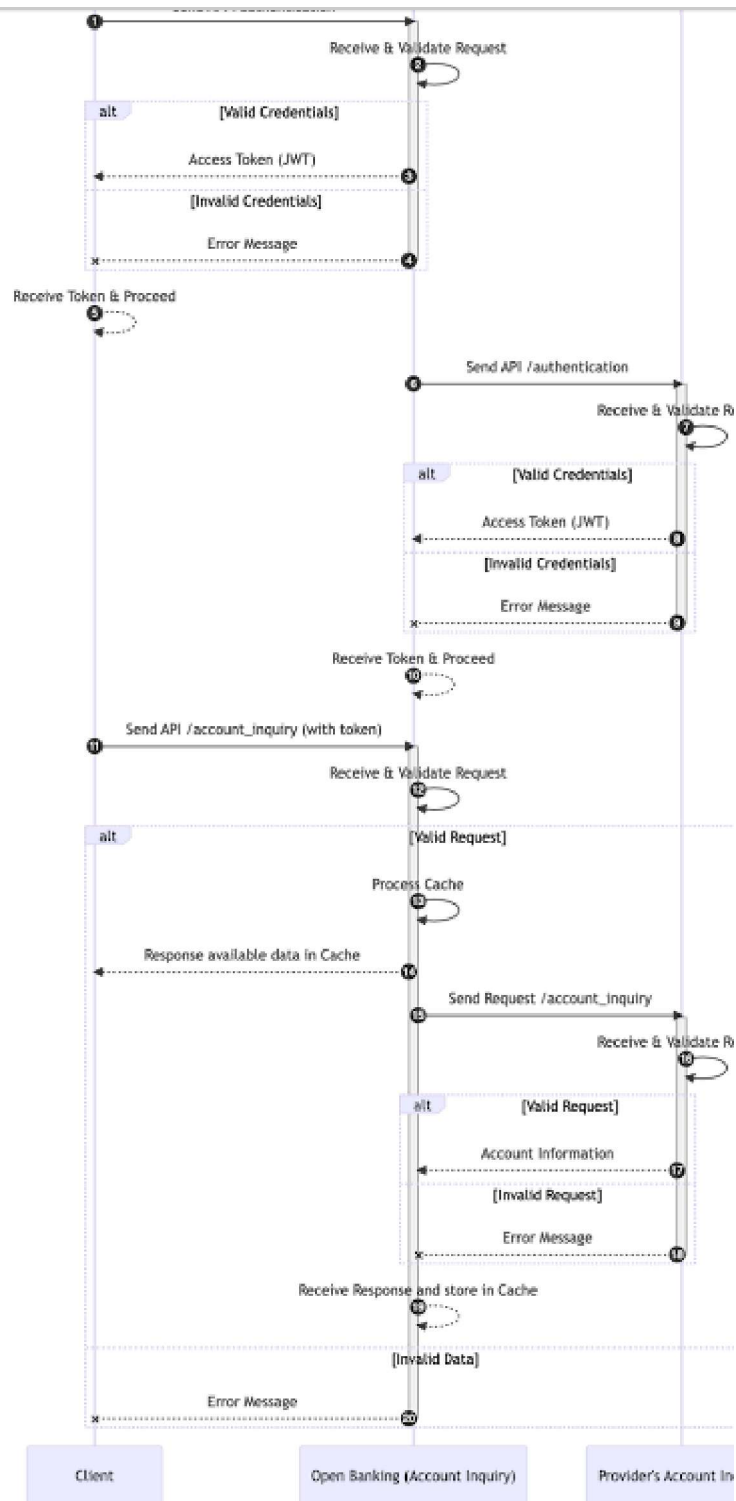


figure 2: Sequence diagram - account inquiry batch

1. [Client Authentication](#)
2. [Provider Authentication](#)
3. Account Inquiry with Token:

← OB - Account Inquiry APIs for client

- The provider validates the request.
 - If valid request: The process proceeds to check the cache.
 - If invalid request: An error message is returned.

4. Cache Check:

- The provider checks its cache for the requested account information.
- If data is in cache: The available data is sent as a response to the client.

5. Account Information Retrieval (If not in cache):

- If data is not in cache: The provider sends a request (`/account_inquiry_batch`) to the provider's account inquiry service.
- The account inquiry service receives and validates the request.
 - If valid request: The account information is returned.
 - If invalid request: An error message is returned.

6. Cache Storage and Response:

- The provider receives the account information (either from the cache or the account inquiry service), stores it in the cache, and sends the response back to the client.
- If invalid data is received from the account inquiry service, an error message is returned to the client.

← OB - Account Inquiry APIs for client

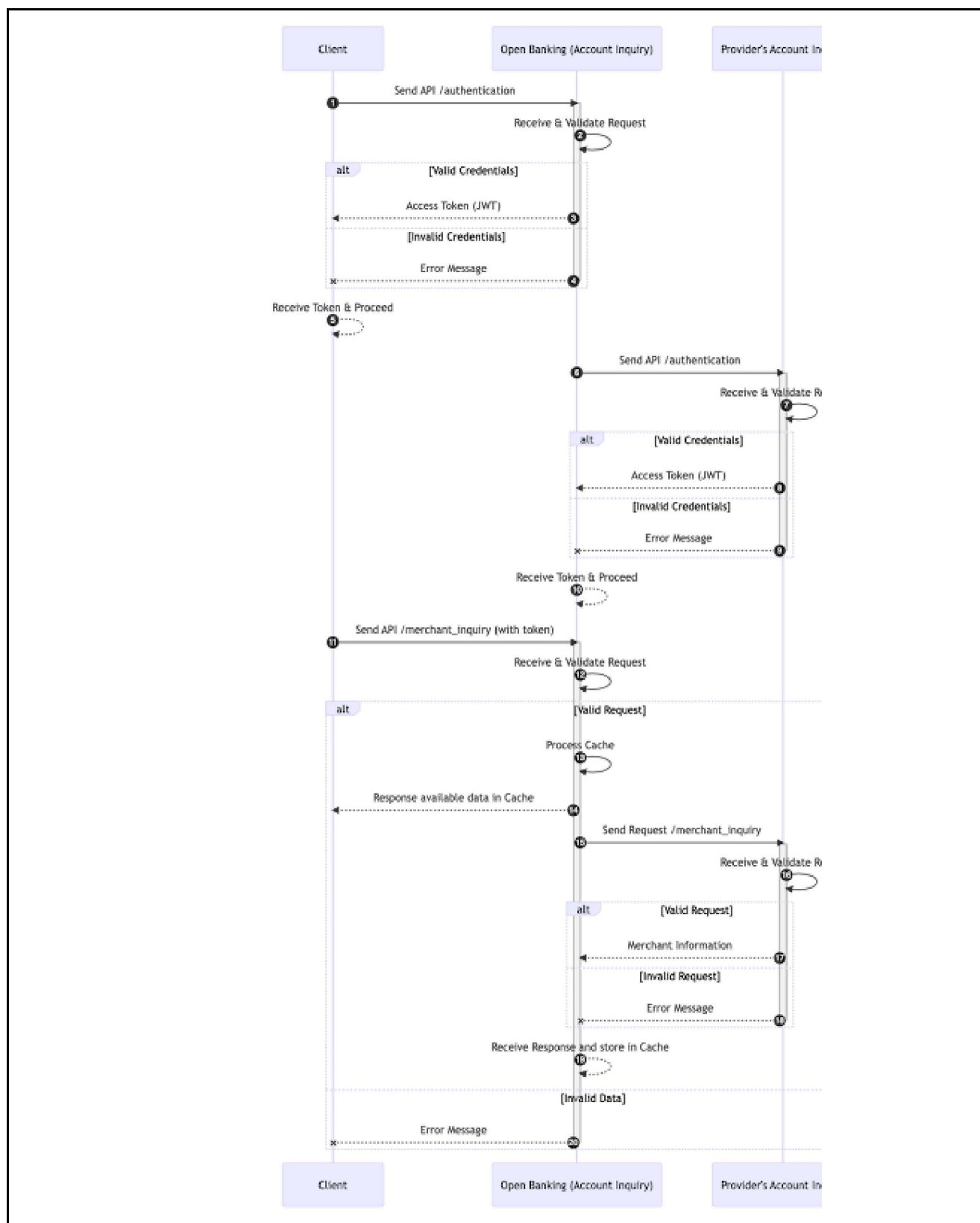


figure 3: Sequence diagram - merchant inquiry

1. [Client Authentication](#)
2. [Provider Authentication](#)

← OB - Account Inquiry APIs for client

- The client sends an account inquiry request ([API/merchant_inquiry](#)) along with the received token to the provider.
- The provider validates the request.
 - If valid request: The process proceeds to check the cache.
 - If invalid request: An error message is returned.

4. Cache Check:

- The provider checks its cache for the requested merchant information.
- If data is in cache: The available data is sent as a response to the client.

5. Account Information Retrieval (If not in cache):

- If data is not in cache: The provider sends a request ([/merchant_inquiry](#)) to the provider's account inquiry service.
- The account inquiry service receives and validates the request.
 - If valid request: The merchant information is returned.
 - If invalid request: An error message is returned.

6. Cache Storage and Response:

- The provider receives the merchant information (either from the cache or the account inquiry service), stores it in the cache, and sends the response back to the client.
- If invalid data is received from the account inquiry service, an error message is returned to the client.

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Main Flow - QR Inquiry

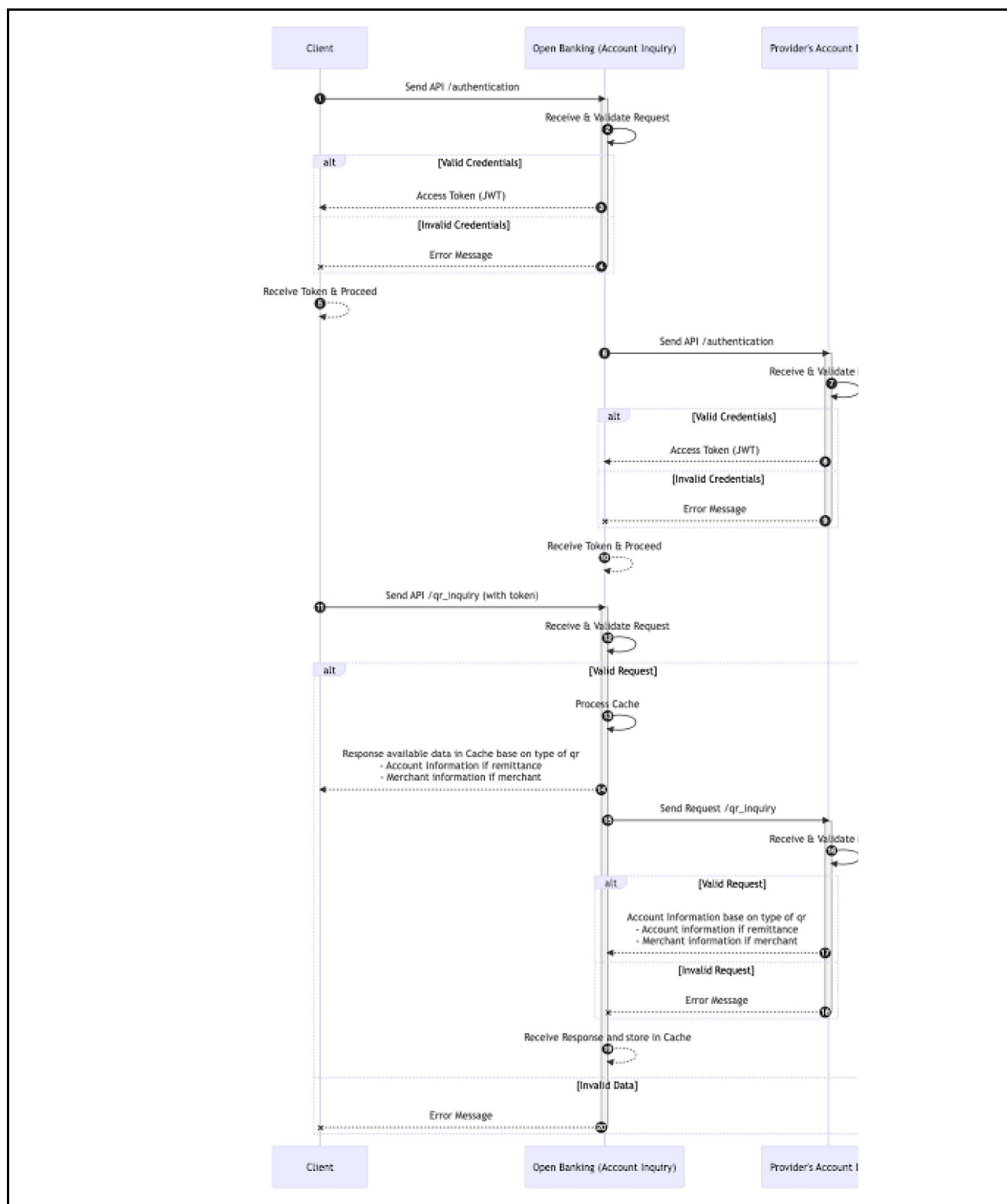


figure 4: Sequence diagram - qr inquiry

← OB - Account Inquiry APIs for client

3. Account/Merchant Inquiry with Token:

- The client sends an inquiry request (**API/qr_inquiry**) along with the received token to the Open Banking provider.
- The Open Banking provider receives and validates the request.
 - If valid request: The process proceeds to check the cache.
 - If invalid request: An error message is returned.

4. Cache Check:

- The Open Banking provider checks its cache for the requested information (account or merchant data).
 - If data is in cache: The available data is sent as a response to the client.

5. Information Retrieval (If not in cache):

- If data is not in cache: The Open Banking provider sends a request (**/qr_inquiry**) to the Provider's Account Inquiry.
- The Provider's Account Inquiry receives and validates the request.
 - If valid request: The requested information (account or merchant) is returned.
 - If invalid request: An error message is returned.

6. Cache Storage and Response:

- The Open Banking provider receives the information (either from the cache or the Provider's Account Inquiry), stores it in the cache, and sends the response back to

← OB - Account Inquiry APIs for client

- If invalid data is received from the Provider's Account Inquiry, an error message is returned to the client.

Appendix Status Code

HTTP Response Code

Code	Text
200	Ok
201	Created
400	Bad request
401	Unauthorized
403	Forbidden
404	Not Found
405	Method not allowed
409	Conflict
411	Length Required
412	Precondition failed
429	Too many requests
500	Internal Server Error
503	Service Unavailable

Custom Error Code

Key	Type	Code	Text
responseCode	Int		

 OB - Account Inquiry APIs for client

errorCode	Int		
		999	Internal Server Error
		998	Recipient Timeout
		997	Service is unavailable
		996	Something went wrong
		1	Authentication error
		2	Validation Error
		3	Invalid QR Code
		4	Invalid amount
		5	Invalid Currency
		6	Invalid Merchant
		7	Invalid Payment Reference
		8	Invalid Credit Account Number
		9	Invalid Transaction
		10	QR Code is not available for payment
		11	QR Code Currency is not support
		12	Merchant over limit amount
		13	Request cannot be processed
		14	Unsupported Country
		15	Account not found
		16	Invalid participant code
		17	QR Expired
		18	QR not found

Response Template

Success

← OB - Account Inquiry APIs for client

```
{  
  "responseMessage": "Authentication successful",  
  "data": {  
    "token": "eyJ0eXAiOiJKV1QiLCJhbGci..."  
  }  
}
```

Failed

```
{
  "responseCode": 1,
  "errorCode": 1,
  "responseMessage": "Authentication error",
  "data": null
}
```

API

Authentication

Note: Clients seeking access to Account Inquiry Module APIs must first obtain an access token through Open Banking's API authentication process.

FIIs need to send a request for the credential from NBC team via email:

sorya.phoeun@nbc.gov.kh

cc: sokhomoliyuth.tea@nbc.gov.kh

HTTP Method	POST
Name	Authentication
Endpoint	{{protocol}}://{{port}}/v1/authenticate
Description	This API is used to obtain an authentication token for authorized Bakong administrators. Once retrieved, the token must be included in the "Authorization" header (often formatted as "Bearer <token>") for all subsequent requests.

Request Parameters

Parameter	Type	Mandatory	Constraint	Description
-----------	------	-----------	------------	-------------

← OB - Account Inquiry APIs for client

Content-Type	String	1		application/json
Body Parameters				
username	String	1		Username used for login.
password	String	1		Password used for login.

Sample Request

```
{
  "username": "username1",
  "password": "username@AA123434",
}
```

Response Parameters

Key		Type	Mandatory	Constraint	Description
responseCode		Integer	1		Code number of the response
errorCode		Integer	1		Code number of the error
responseMessage		String	1		Description of the result
data		Object	1		Data of the response
	token	String	1	lifetime - min:1 day - max: 30 days	JWT token, Bearer {{token}}

Sample Response

```
{
  "responseCode": 0,
  "errorCode": null,
  "responseMessage": "Authentication successful",
  "data": {
    "token": "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJ1b250b29udGVudCI6ImF1dG8iLCJ0eXBlIjoiYXV0b29udGVudCJ9.eyJ1b250b29udGVudCI6ImF1dG8iLCJ0eXBlIjoiYXV0b29udGVudCJ9"
  }
}
```

 OB - Account Inquiry APIs for client

Account Inquiry

HTTP Method	POST
Name	Account Inquiry
Endpoint	{{protocol}}://{{host}}/api/v1/account_inquiry/account
Description	Retrieve information about a customer's specific account

Request Parameters

Parameter	Type	Mandatory	Description
Header parameters			
Content-Type	String	1	application/json
Authorization	String	1	Bearer {{token}}
App-Name	String	1	BANK {{client participant code}} (sender bank as client that request to provider) Ex: BANK ACLBKHPPXXX Or BAKONG
Local-Part-Code	String	1	Provider participant code (destination bank that client requested) Ex: ADEOKHPPXXX
Type	String	1	Value: - LOCAL (by default) : Local transfer - INTERNATIONAL (coming soon) : International transfer
Body Parameters			
accountNumber	String	1	The bank or FI assigns a unique identifier to a customer's financial account Note: - If App-Name = BAKONG: accountNumber can be either : bakong account id or primary contact. Ex: test@devb OR 85512123456 - If App-Name = BANK {{client participant code}} : accountNumber = CASA Account Number

← OB - Account Inquiry APIs for client

Sample Request

```
{
  "accountNumber": "85512777661" //test@devb //123456789
}
```

Response Parameters

Key	Type	Mandatory	Description
errorCode	Integer	1	The code number of the error
responseCode	Integer	1	The code number of the response
responseMessage	String	0	Description of the result
data	object		
accountInfo	Object	1	The object of the account information
name	String	1	Account name
accountNumber	String	1	Account Number
localPartCode	String	1	Bank code
foreignPartCode	String	0	Foreign bank code for international transfer
ccy	Array	1	Currency of account Available values: - USD - KHR
type	String	1	Account type Available values: - SAVINGS - CURRENT - WALLET - LOAN *Note: For other account types, please inform the NBC team.
openingDate	String	1	Date the bank activated the account Available format: - dd/MM/yyyy
status	String	1	Current operational account state Available values: - ACTIVATED

← OB - Account Inquiry APIs for client

		kycStatus	String	1	Know Your Customer account status Available values: - FULL_KYC - PARTIAL_KYC
		accountId	String	0	Unique identifier for the Bakong wallet account.
		primaryContact	String	0	The primary contact associated with the account, which could be an email or phone number.
		frozen	Boolean	0	Indicates whether the account is frozen.

Sample Response

- Successful Responses - account status is ACTIVE

```
{
  "errorCode": null,
  "responseCode": 0,
  "responseMessage": "Getting account information successfully",
  "data": {
    "accountInfo": {
      "name": "Ty Bopha",
      "accountNumber": "000123456",
      "localPartCode": "DEVBKHPPXXX",
      "foreignPartCode": "CZCHXXX",
      "ccy": [
        "USD",
        "KHR"
      ],
    },
    "type": "WALLET",
    "openingDate": "23/01/2023",
    "status": "ACTIVE",
    "kycStatus": "FULL_KYC",
    "accountId": "jonh@devb",
    "primaryContact": "85512345678",
  }
}
```

← OB - Account Inquiry APIs for client

```
}
```

- Successful response for account status is INACTIVE

```
{
  "errorCode": null,
  "responseCode": 0,
  "responseMessage": "Getting account information successfully",
  "data": {
    "accountInfo": {
      "name": "Ty Bopha",
      "accountNumber": "0001234567",
      "localPartCode": "DEVBKHPPXXX",
      "foreignPartCode": "CZCHXXX",
      "ccy": [
        "USD",
        "KHR"
      ],
      "type": "WALLET",
      "openingDate": "23/01/2023",
      "status": "INACTIVE",
      "kycStatus": "FULL_KYC",
      "accountId": "jonh@devb",
      "primaryContact": "85512345678",
      "frozen": false
    }
  }
}
```

- Account not found

```
{
  "errorCode": 15,
  "responseCode": 1,
  "responseMessage": "Account not found",
  "data": null
}
```

Account Inquiry Batch

HTTP Method	POST
Name	Account Inquiry Batch
Endpoint	{{protocol}}://{{host}}/api/v1/account_inquiry/account_inquiry_batch

← OB - Account Inquiry APIs for client

Request Parameters

Parameter	Type	Mandatory	Description
Header parameters			
Content-Type	String	1	application/json
Authorization	String	1	Bearer {{token}}
App-Name	String	1	BANK {{client participant code}} (sender bank as client that request to provider) Ex: BANK ACLBKHPPXXX
Local-Part-Code	String	1	Provider participant code (destination bank that client requested) Ex: ADEOKHPPXXX
Type	String	1	Value: - LOCAL (by default) : Local transfer - INTERNATIONAL (coming soon) : International transfer
Body Parameters			
accountNumbers	Array	1	The bank or FI assigns unique identifiers to a customer's financial account. Note: - Max 100 accounts

Sample Request

```
{
  "accountNumbers": [
    "85512777661",
    "00852518",
    "008525790"
  ]
}
```

Response Parameters

Key	Type	Mandatory	Description
errorCode	Integer	1	The code number of the error

← OB - Account Inquiry APIs for client

responseMessage		String	0	Description of the result
data		Objects	1	The list data of the response
	accountInfo	Array of objects	1	Array of the account information
	accountNumber	String	1	The bank or financial institution assigns a unique identifier to a customer's financial account
	name	String	1	Account name
	localPartCode	String	1	Bank code
	foreignPartCode	String	0	Foreign bank code for international transfer
	ccy	Array	1	Currency of account Available values: - USD - KHR
	type	String	1	Account type Available values: - SAVINGS - CURRENT - WALLET - LOAN *Note: For other account types, please inform the NBC team.
	openingDate	String	1	Date the bank activated the account Available format: - dd/MM/yyyy
	status	String	1	Current operational account state Available values: - ACTIVATED - DEACTIVATED (meaning block or frozen and dormant)
	kycStatus	String	1	Know Your Customer account status Available values: - FULL_KYC - PARTIAL_KYC
accountNotFound		Array of objects	1	The list of account numbers that were not found

Sample Response

← OB - Account Inquiry APIs for client

```

responseCode : 0,
"responseMessage": "Getting account information successfully",
"data": {
  "accountInfo": [
    {
      "accountNumber": "85512777661",
      "name": "Ty Bopha",
      "localPartCode": "DEVBKHPPXXX",
      "foreignPartCode": null,
      "ccy": [
        "USD",
        "KHR"
      ],
      "type": "WALLET",
      "openingDate": "23/01/2023",
      "status": "ACTIVE",
      "kycStatus": "FULL_KYC"
    },
    {
      "accountNumber": "00852518",
      "name": "Ty Bopha",
      "localPartCode": "DEVBKHPPXXX",
      "foreignPartCode": null,
      "ccy": [
        "USD"
      ],
      "type": "SAVING",
      "openingDate": "23/01/2023",
      "status": "ACTIVE",
      "kycStatus": "FULL_KYC"
    }
  ],
  "accountNotFound": [
    "008525790"
  ]
}
}

```

Merchant Inquiry (Optional)

HTTP Method	POST
Name	Merchant Inquiry
Endpoint	{{protocol}}://{{host}}/api/v1/account_inquiry/merchant_inquiry
Description	Retrieve information about a merchant associated with a transaction

← OB - Account Inquiry APIs for client

Parameter	Type	Mandatory	Description
Header parameters			
Content-Type	String	1	application/json
Authorization	String	1	Bearer {{token}}
App-Name	String	1	BANK {{client participant code}} (sender bank as client that request to provider) Ex: BANK ACLBKHPPXXX
Local-Part-Code	String	1	Provider participant code (destination bank that client requested) Ex: ADEOKHPPXXX
Body Parameters			
merchantId	String	1	Merchant ID

Sample Request

```
{  
  "merchantId": "256784422"  
}
```

Response Parameters

Key			Type	Mandatory	Description
responseCode			Integer	1	Please follow the custom code
errorCode			Integer	1	Please follow the custom code
responseMessage			String	0	Please follow the custom code
data			Object	1	
	merchantInfo		Object	1	The object of the merchant
		name	String	1	The official business name. This is the most recognizable piece of information for the customer
		terminalId	String	0	Terminal ID such as POS, ATM, Kiosk, and Mobile Payment

← OB - Account Inquiry APIs for client

				is a four-digit code classifying the business type. It affects fees, rewards, and tax reporting.
	localPartCode	String	1	Bank code
	type	String	1	
	status	String	1	
	kycStatus	String	1	
	openingDate	String	1	
	address	Object		
	street	String	1	
	city	String	1	The city or town where the business is located. This is helpful for deliveries and finding nearby stores
	province	String	1	
	postalCode	String	1	
	country	String	1	
	ccy	Array	1	Currency <u>Available values:</u> - USD - KHR

Sample Response

- Successful response

```
{
  "errorCode": null,
  "responseCode": 0,
  "responseMessage": "Getting merchant information successfully",
  "data": {
    "merchantInfo": {
      "name": "Brown",
      "terminalId": "T12345",
      "mcc": "5999",
      "localPartCode": "CSXXX",
      "type": "CURRENT",
      "status": "ACTIVE",
      "kycStatus": "FULL_KYC",

```

← OB - Account Inquiry APIs for client

```

    "city": "Phnom Penh",
    "province": "Phnom Penh",
    "postalCode": "120101",
    "country": "Cambodia"
  },
  "ccy": [
    "USD",
    "KHR"
  ]
}
}
}

```

- Invalid merchant

```

{
  "errorCode": 6,
  "responseCode": 1,
  "responseMessage": "Invalid merchant",
  "data": null
}

```

QR Inquiry

HTTP Method	POST
Name	QR Inquiry
Endpoint	{{protocol}}://{{host}}/api/v1/account_inquiry/qr_inquiry
Description	Retrieve information encoded within a QR code used for payments. NOTE: QR can be remittance or merchant

Request Parameters

Parameter	Type	Mandatory	Description
Header parameters			
Content-Type	String	1	application/json
Authorization	String	1	Bearer {{token}}
App-Name	String	1	BANK {{client participant code}} (sender bank as client that request to provider)

← OB - Account Inquiry APIs for client

bank that client requested)
Ex: ADEOKHPPXXX

Body Parameters

qrContent	String	1	Qr content string Available values: - Remittance QR - Merchant QR
-----------	--------	---	---

Sample Request

```
{
  "qrContent": "eyJhbGciOiJIUzUxMiJ9nmdsfgd"
}
```

Response Parameters

Key		Type	Mandatory	Description	
responseCode		Integer	1	Please follow the custom code	
errorCode		Integer	1	Please follow the custom code	
responseMessage		String	0	Please follow the custom code	
data		Object	1	data response based on QR type <u>NOTE:</u> - remittance, response accountInfo - merchant, response merchantInfo	
	QRInfo		Object	1	The object of the account information from QR-type remittance
		name	String	1	Account name
		ccy	Array	1	Currency of account <u>Available values:</u> - USD - KHR
		type	String	1	Account type <u>Available values:</u> - SAVING - CURRENT - WALLET - LOAN

← OB - Account Inquiry APIs for client

		status	String	1	Current operational account state Available values: - ACTIVE - INACTIVE (meaning block or frozen and dormant)
		kycStatus	String	1	Know Your Customer account status Available values: - FULL_KYC - PARTIAL_KYC
		merchantId	String	1	Merchant ID that represents ID to credit amount to merchant
		terminalId	String	0	Terminal ID such as POS, ATM, Kiosk, and Mobile Payment Terminal.
		mcc	String	1	(MCC) is a four-digit code classifying the business type. It affects fees, rewards, and tax reporting.
		city	String	0	The city or town where the business is located. This is helpful for deliveries and finding nearby stores

Sample Response (remittance)

- Success response - remittance tag 29

```
{
  "errorCode": null,
  "responseCode": 0,
  "responseMessage": "Getting QR information successfully",
  "data": {
    "QRInfo": {
      "name": "Ty Bopha",
      "ccy": [
        "KHR"
      ],
      "type": "SAVINGS",
      "status": "ACTIVE",
      "kycStatus": "FULL_KYC",
      "merchantId": null,
      "terminalId": null,
      "mcc": null,
      "city": null
    }
  }
}
```

← OB - Account Inquiry APIs for client

- Success response - merchant tag 30

```
{
  "errorCode": null,
  "responseCode": 0,
  "responseMessage": "Getting QR information successfully",
  "data": {
    "QRInfo": {
      "merchantId": "256784422",
      "terminalId": "T12345",
      "name": "Brown",
      "mcc": "CS001",
      "city": "Phnom Penh",
      "ccy": [
        "USD",
        "KHR"
      ],
      "type": null,
      "status": null,
      "kycStatus": null
    }
  }
}
```

- Invalid QR

```
{
  "errorCode": 3,
  "responseCode": 1,
  "responseMessage": "Invalid QR Code",
  "data": null
}
```

- QR Expired

```
{
  "errorCode": 17,
  "responseCode": 1,
  "responseMessage": "QR Expired",
  "data": null
}
```

- QR not found

```
{
  "errorCode": 18,
  "responseCode": 1,
  "responseMessage": "QR not found",
}
```


← OB - Account Inquiry APIs for client

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